DSA 3050A – Business Intelligence & Data Visualization

Practical Lab: DAX Using Your Own Dataset

Lab Submission Report

1. Dataset Description

Dataset Name: USA Banking Transactions Dataset (2023–2024)

Source: https://www.kaggle.com/datasets/pradeepkumar2424/usa-banking-transactions-dataset-2023-2024

This dataset contains banking transaction records from the United States covering the period 2023 to 2024. It includes details such as transaction amounts, transaction types, account categories, payment methods, and geographic cities. Each record is timestamped, making it suitable for time-based and trend analysis. The dataset was sourced from Kaggle and contains 5,389 rows of real-world financial transaction data. It was used to analyse transaction patterns, fraud detection, regional performance, customer demographics, and year-over-year financial trends.

Column	Type	Description
Transaction_Date	Date	Date of the transaction
Transaction_Amount	Decimal	Value of the transaction in USD
Transaction_Type	Text	Debit or Credit
Category	Text	Spending category (e.g. Travel, Grocery)
City	Text	City where transaction occurred
Payment_Method	Text	Method used (e.g. Cash, E-Wallet)
Transaction_Status	Text	Success, Failed, or Pending
Fraud_Flag	Text	Whether transaction was flagged as fraud
Customer_Age	Whole Number	Age of the customer
Customer_Income	Decimal	Annual income of the customer
Account_Balance	Decimal	Account balance at time of transaction
Loyalty_Points_Earned	Whole Number	Points earned per transaction
Discount_Applied	Text	Whether a discount was applied

2. DAX Measures Explained

Section B – Core Measures

Measure Name	DAX Formula	What It Calculates
Total Transaction Amount	SUM(Transactions[Transaction_Amount])	Adds up all transaction amounts across the dataset
Total Customer Income	SUM(Transactions[Customer_Income])	Sums up the annual income of all customers in context
Average Transaction Amount	AVERAGE(Transactions[Transaction_Amount])	Calculates the mean transaction value dynamically
Total Loyalty Points	SUM(Transactions[Loyalty_Points_Earned])	Total loyalty points earned across all transactions

Section C – Calculated Columns

Column Name	Logic	Purpose
Transaction Size	IF Amount < 500 = Small, < 2000 = Medium, else Large	Classifies each transaction into a size band for grouping
Age Group	IF Age < 30 = Young, < 50 = Middle Aged, else Senior	Groups customers by age bracket for demographic analysis

Section D – CALCULATE Measures

Measure Name	What It Does
Successful Transactions Amount	Uses CALCULATE to filter only transactions with Status = Success, ignoring failed and pending ones
Fraud Transactions Amount	Uses CALCULATE to isolate transactions where Fraud_Flag = Yes, showing total fraudulent value
% Category Contribution	Uses CALCULATE with ALL() to remove the category filter and compute each category's share of the grand total

Section E – Iterator Function

Measure Name	What It Does
Revenue After Discount	Uses SUMX to loop row by row — if a discount was applied, it multiplies the amount by 0.9 (10% off); otherwise it keeps the original amount. A regular SUM cannot do this row-level conditional logic.

Section F – Time Intelligence

Measure Name	What It Does
YTD Transaction Amount	Uses TOTALYTD to accumulate transaction amounts from January 1st to whatever date is selected in the slicer

Measure Name	What It Does
Previous Year Amount	Uses PREVIOUSYEAR to shift the date context back 12 months, showing last year's total for comparison
YoY Growth %	Divides the difference between current and previous year by the previous year amount to show percentage growth or decline

3. One Challenge Encountered

The main challenge encountered during this lab was the Transaction_Date column. When the dataset was first loaded into Power BI, the column was recognised as a DateTime format (e.g. 2023-11-05 15:54:38) rather than a plain Date. This caused the relationship between the Transactions table and the DateTable to fail silently — the time intelligence functions returned blank results because the date keys could not match properly.

The issue was resolved by opening Power Query Editor and manually changing the Transaction_Date column type from DateTime to Date. This stripped the time component and allowed Power BI to establish a clean one-to-many relationship between the DateTable and the Transactions table. After this fix, all time intelligence measures including YTD, Previous Year, and YoY Growth began returning correct results and responded properly to date slicers.

4. Visuals Summary

Visual Type	Fields Used	Purpose
Card	Total Transaction Amount	Shows the overall total as a single headline number
Bar Chart	Category vs Total Transaction Amount	Compares spending across categories like Travel, Shopping, Healthcare
Line Chart	YearMonth (DateTable) vs Total Transaction Amount	Shows transaction trends month by month across 2023–2024
Table	Category, Total Amount, Successful Amount, Fraud Amount, % Contribution	Provides a detailed breakdown per category in one view